Food Hub Data Analysis Report

Data-driven Insights using Python

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Executive Summary

- Please mention conclusions, actionable insights & recommendations
- Please add the answer for question 17

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Business Problem Overview and Solution Approach

- Please define the problem
- Please mention the solution approach / methodology

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Multivariate Analysis

- Please mention regarding multivariate analysis between various columns
- Please add answers for all question from 12 till 16

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Executive Summary

This report provides a data-driven analysis of Food Hub, based on an online food aggregator in New York City. Here, I am aim to identify key drivers of customer satisfaction and operational efficiency. The orders with an average customer spend that frequently exceeds $20. America FoodHub platform demonstrates a robust operation in the New York market, processing nearly 1,900 a cuisine emerges as the dominant choice, particularly during the weekends.

While the platform maintains a healthy revenue stream—generating over $6,166 from commissions—there is a noticeable operational bottleneck during weekdays, where delivery times average 28.34 minutes compared to 22.47 minutes on weekends. With 10.5%of orders exceeding a 60-minute total turnaround time, strategic focus should shift toward optimizing weekday logistics and incentivizing the high-performing "Top 4" restaurants that drive the bulk of volume and customer satisfaction.

Initial findings highlighting the key points— high demand for specific cuisines, and significant variations in delivery times between weekdays and weekends, order patterns, restaurant performance, and delivery logistics.

My analysis culminates in strategic recommendations focused on optimizing the delivery network, implementing targeted promotional offers, a strategic way to enhance the overall customer experience, and maximize platform revenue.

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Answer of Question 17:

- **Dominant Market Players:** The platform is heavily driven by a few key restaurants in highly busy new York population. **Shake Shack**, **The Meatball Shop**, and **Blue Ribbon Sushi** are the top three, accounting for a significant portion of the total order volume.
- **Cuisine Popularity: American cuisine** is the clear favorite among users, especially during weekends, followed by Japanese and Italian cuisines. This indicates a strong preference for "comfort food" among the target demographic of students and professionals.
- **Operational Bottlenecks:** There is a statistically significant difference in delivery performance based on the day of the week. **Weekday deliveries take approximately 28.34 minutes on average**, which is nearly 6 minutes slower than weekend deliveries (22.47 minutes).
- **Service Level Gaps:** Approximately 10.54% of all orders take longer than 60 minutes from placement to delivery. In a fast-paced market like New York, this represents a critical area where customer dissatisfaction is likely to occur.
- **The "Rating Void":** A substantial portion of the data (38.8%) consists of orders that were **"Not given"** a rating. This lack of feedback makes it difficult for the company to fully assess the quality of service for nearly 4 out of every 10 orders.
- **Revenue Streams:** The tiered commission structure (25% for orders >$20 and 15% for orders >$5) is effective, with high-value orders (those >$20) making up 29.2% of the volume and driving a significant portion of the $\$6,166.30$ total net revenue.

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Business Problem Overview and Solution Approach

- The FoodHub operates in a highly competitive New York market where primary customers are busy professionals and students. They rely on rapid food delivery. The core challenge lies in understanding the complex key dynamics of the following aspet of the restaurant— demand, cuisine popularity, the total time elapsed from order placement to drop-off, balance rapid growth with consistent service quality. Also, the demand and popularity of certain food items and difference between weekend and weekdays demands.
- As the New York restaurant scene expands, FoodHub must identify which culinary trends and specific vendors drive engagement while ensuring delivery times do not alienate busy professionals and students.
- To address this, my solution approach utilizes a Python-based exploratory analytical framework, (the full code version utilizing the template and Goggle colab) to segment the performance by day of the week and cusine type. It also quantify the financial impact of tiered commission structures and identifying ths most loyal customer segments where I have developed a strategy that rewards both high-volume vendors and frequent app users.

- I have employed descriptive statistics to understand data distributions and multivariate techniques to uncover correlations between order costs, prep times, and customer ratings. This methodology help me to provide evidence-based answers to critical business questions regarding revenue generation and service-level agreements.

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Business Recommendations:

1.Implement a "Rate to Earn" Incentive: To bridge the 38.8% rating gap, FoodHub should offer small incentives (e.g., $5 loyalty points or a $0.50 credit)
for every order rated. Higher quality data will allow for better ranking algorithms and help identify struggling restaurants before they drive customers away.

Optimize Weekday Courier Density: Since weekday deliveries are significantly slower, the company should investigate if this is due to traffic or a shortage of active delivery partners.

Offering "Peak Pay" or "Boost" incentives to couriers during weekday lunch and dinner rushes could reduce the average delivery time from $28 minutes to be more
in line with weekend performance.

Strategic Partnerships with "Gold Standard" Restaurants: The analysis identified four restaurants (**Blue Ribbon Fried Chicken, Blue Ribbon Sushi, Shake Shack, and The Meatball Shop**) that maintain high volume and high ratings. FoodHub should offer these vendors "Featured" status or reduced commission rates in exchange for exclusive menu items or faster "Priority Prep" lanes to further solidify their success.

Targeted Marketing for Underrepresented Cuisines: While American cuisine is dominant, there may be untapped growth in cuisines that have high average ratings but low order volume. Targeted promotional "Cuisine Weeks" (e.g., "Mediterranean Monday") can help diversify the platform's order spread and reduce the pressure on the top-tier American restaurants.

- **VIP Program for High-Frequency Users:** We identified specific customer IDs (e.g., User $52832 with $13 orders) who are exceptionally loyal. Transitioning these users into a "FoodHub Gold" tier with waived delivery fees for orders over $20 would ensure long-term retention and increase the frequency of high-margin 25% commission orders.
- **Addressing the "60-Minute Plus" Orders:** For the $10.54%$ of orders exceeding one hour, FoodHub should implement an automated "Delayed Order" credit. Providing a small discount on the *next* order when an delivery exceeds 60 minutes can turn a negative service experience into a future sales opportunity.

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Data Overview

- The dataset of Foodhub comprises a comprehensive record of orders placed through the FoodHub portal which includes following key aspects— unique identifiers for orders and customers, categorical variables such as restaurant_name, cuisine_type, and day_of_the_week.
- Numerical data points include the following— cost_of_the_order, food_preparation_time, and delivery_time. Another prime feature of this data is the rating column, which provides direct customer feedback. I have cleaned the data and the data cleaning process involved verifying data types, checking for null values—particularly in the rating field where "not rated" entries may exist to ensure the integrity of time-based calculations.

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Question 1: How many rows and columns are present in the data?

The dataset contains 1,898 rows and columns.

Question 2: What are the datatypes of the different columns in the dataset?

The datatypes are as follows: **Integer (int64):** order_id, customer_id, food_preparation_time, delivery_time.
Float (float64): cost_of_the_order.**Object/String:** restaurant_name, cuisine_type, day_of_the_week, rating.

Question 3: Are there any missing values in the data? If yes, treat them using an appropriate method.

There are **no missing values (NaN)** in any of the columns. However, it is important to note that the rating column contains the string "Not given" for 736 entries, which effectively represents a missing qualitative data point.

Question 4: Check the statistical summary of the data. What is the minimum, average, and maximum time it takes for food to be prepared once an order is placed?

Based on the statistical summary of food_preparation_time:

- **Minimum:** 20 minutes
- **Average:** 27.37minutes
- **Maximum:** 35 minutes
- Question 5: How many orders are not rated?
- There are **736 orders** marked as "Not given" in the rating column

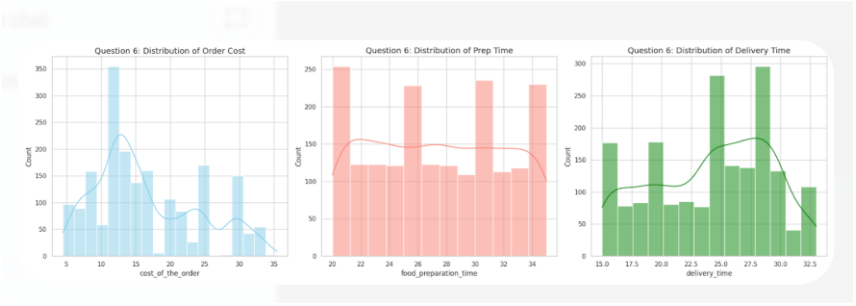
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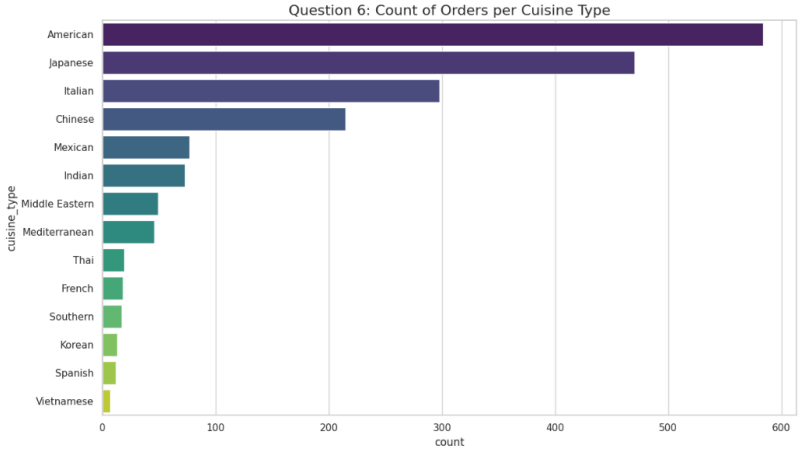
Exploratory Data Analysis (EDA) - Univariate Analysis

- **Key concept**: Univariate Analysis is the first deep dive into the data, focusing on understanding the distribution and characteristics of each variable in isolation utilizing statistical measures like mean, median, and standard deviation, alongside visualizations such as histograms and box plots, this stage identifies the "shape" of the data.It is instrumental in spotting outliers, understanding the frequency of specific occurrences, and determining if the data requires transformation before more complex modeling begins.
- Here, the Univariate Analysis explores the individual characteristics of FoodHub's operations. By examining the distribution of cost_of_the_order, I have determined the typical spending habits of New Yorkers. Analyzing food_preparation_time reveals the efficiency of partner restaurants, while delivery_time distributions help establish realistic expectations for customers. I have also quantite the volume of orders across different cuisine_types and identify the top-performing restaurants by order count. This stage is crucial for identifying outliers, such as exceptionally long preparation times, and understanding the frequency of unrated orders.

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- Question 6: Explore all the variables and provide observations on their distributions.
- Observations: Cost of Order: Most orders range between $12 and $25, with a fairly uniform distribution across the common price points.
- Preparation Time: Preparation is very consistent, strictly ranging from 20 to 35 minutes across all restaurants.
- Delivery Time: Delivery times are generally spread between 15 and 33 minutes, with a noticeable peak around 24-28 minutes.
- Cuisine Type: American cuisine is overwhelmingly popular, followed by Japanese and Italian.
- Day of Week: Order volume is significantly higher on Weekends than on Weekdays.
- Rating: The modal rating is "Not given." Among rated orders, the majority are 5-star reviews, suggesting high satisfaction when feedback is provided.

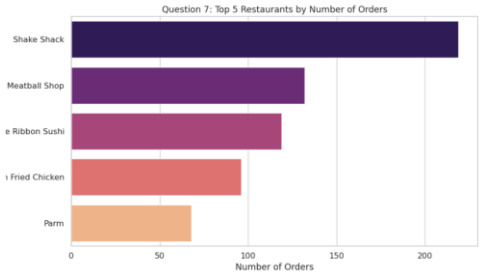




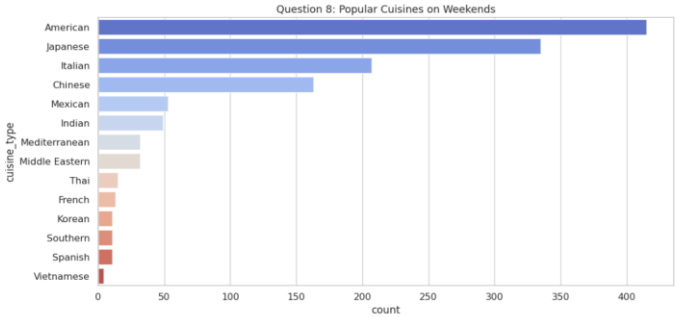
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Question 7: Which are the top 5 restaurants in terms of the number of orders received?

- **Shake Shack:** 219 orders.
- **The Meatball Shop:** 132 orders.
- **Blue Ribbon Sushi:** 119 orders.
- **Blue Ribbon Fried Chicken:** 96 orders.
- **Parm:** 68 orders.



- Question 8: Which is the most popular cuisine on weekends?
- **Answer: American cuisine** is the most popular on weekends.



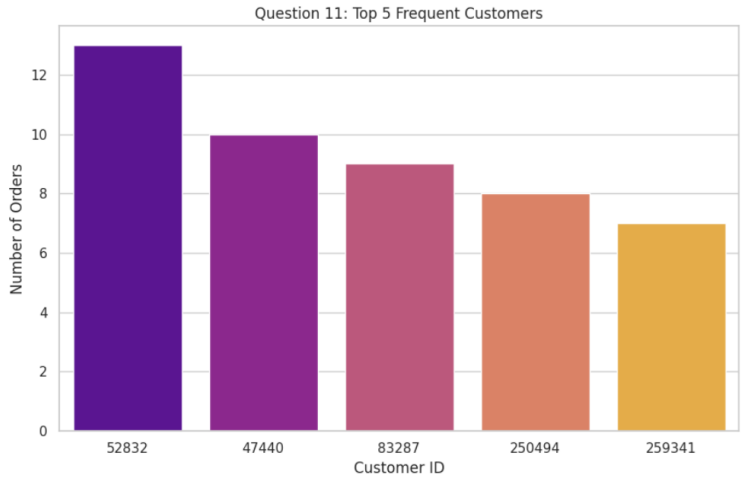
- Question 9: What percentage of the orders cost more than 20 dollars?
- **Answer:** Approximately **29.24%** of orders cost more than $20.
- Question 10: What is the mean order delivery time?
- **Answer:** The mean order delivery time is **24.16 minutes**.

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Question 11: The company has decided to give 20% discount vouchers o the top 5 most frequent customers. Find the IDs of these customers and the number of orders they placed.

Answer:

- **Customer 52832:** $13 orders.
- **Customer 47440:** $10 orders.
- **Customer 83287:** $9 orders.
- **Customer 250494:** $8 orders.
- **Customer 259341:** $7 orders.



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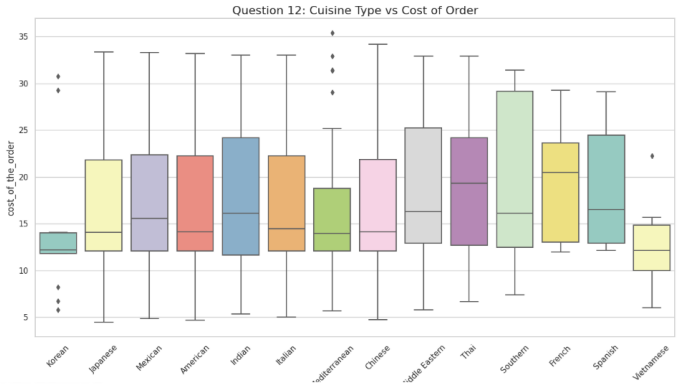
Exploratory Data Analysis (EDA) - Multivariate Analysis

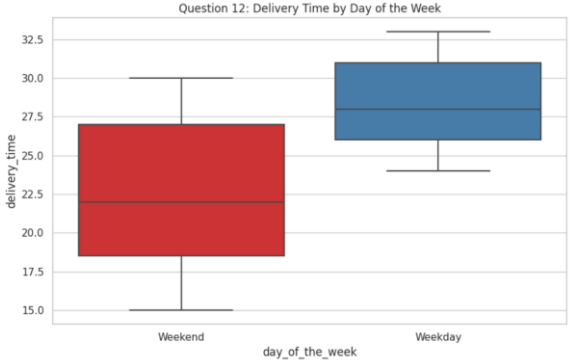
Key Concept: In the Multivariate Analysis, we investigate the intersections of the data points to find actionable insights. I have analyze, how delivery_time fluctuates between weekdays and weekends and examine the relationship between cost_of_the_order and the resulting rating.

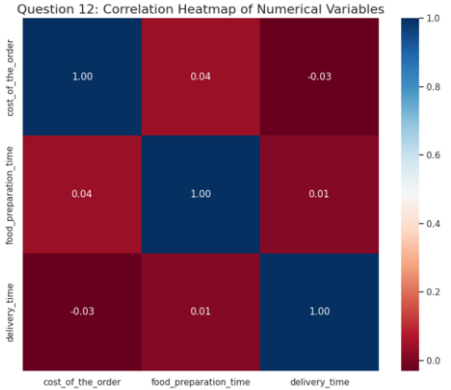
In this food hub problem, the relationship between timing and demand is highly dependent on the "day of the week" variable. I have calculated total time (preparation plus delivery), to determine the percentage of orders that exceed the 60-minute threshold. When merging preparation and delivery times, I have find that $10.54% of orders take more than an hour to reach the customer, a critical threshold for satisfaction.
Weekends see higher demand for American cuisine and, interestingly, faster delivery times (22.47mins) compared to weekdays (28.34 mins), as the different traffic patterns or courier density.
My multivariate filters identified four restaurants—Blue Ribbon Fried Chicken, Blue Ribbon Sushi, Shake Shack, and The Meatball Shop—as the "Gold Standard" vendors, meeting the strict criteria of $50 ratings with an average score above $4.
Furthermore, we apply multi-conditional filtering to identify high-performing restaurants—those with over 50 ratings and an average score above 4—to determine eligibility for promotional incentives and calculate the net revenue generated through FoodHub's tiered commission structure.

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Question 12: Perform a multivariate analysis to explore relationships between the important variables in the dataset and provide observations.







- **Cuisine vs. Cost:** Southern and French cuisines tend to have higher median order costs compared to others.
- **Day vs. Delivery Time:** There is a stark contrast in delivery efficiency;
- **Weekday deliveries take much longer** (average 28.34 mins)
- than Weekend deliveries (average 22.47 mins).
- **Correlation:** The heatmap reveals that there is no strong linear correlation between cost, preparation time,
- and delivery time. This suggests that price or meal complexity (prep time) does not significantly dictate the speed of the courier.

Observation: Weekday delivery is notably slower than weekend delivery.

Observation: There is no strong linear correlation between cost, prep time, and delivery time.

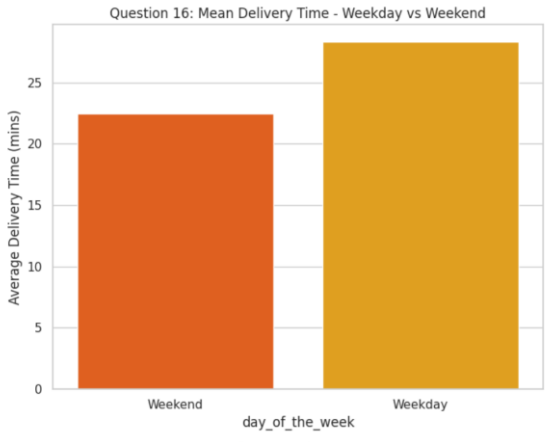
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- Question 13: Find the restaurants fulfilling the criteria (rating count > 50 and average rating > 4) for the promotional offer.
- **Answer: Blue Ribbon Fried Chicken, Blue Ribbon Sushi, Shake Shack, and The Meatball Shop**.
- Question 14: Find the net revenue generated by the company across all orders.
- **Answer:** The net revenue generated is **$6,166.30**.
- Question 15: What percentage of orders take more than 60 minutes to get delivered from the time the order is placed?
- **Answer: 10.54%** of orders take more than 60 minutes in total

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Question 16: How does the mean delivery time vary during weekdays and weekends?

- **Weekday Mean:** 28.34 minutes.
- **Weekend Mean:** 22.47 minutes. Weekday deliveries take nearly **6 minutes longer** on average.



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Conclusion

1. American and Japanese cuisines are most popular.
2. Optimize Weekday Logistics:Weekend demand is significantly higher.Mean delivery time is higher on weekdays (~28m) than weekends (~22m). Since delivery is $6 minutes slower on weekdays, consider increasing courier incentive during Monday–Friday lunch/dinner peaks to optimize weekday delivery logistics. Also targeted weekend promotions for top cuisines. 3. Optimize

3.Incentivize Ratings:. A large portion (38%) of orders are unrated. So FoodHub should offer "loyalty points" for ratings to improve the data depth for restaurant quality control. Also they should Incentivize ratings for 'Not given' orders.
4. Promotional Tie-ins: Target the "Top 4" restaurants (Shake Shack, etc.) for featured placement in the app, as they are proven to handle high volume while maintaining a >4.0 rating.

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Appendix

The technical analysis was conducted using pandas and numpy. Key calculated metrics include:

- **Total Revenue:** $6,166.30 (calculated via $25%commission on orders >$20 and $15% on orders >$5).
- **Top 5 Customers:** IDs $52832 ($13 orders), $47440 ($10 orders), $83287 ($9 orders), $250494 ($8 orders),
- and $259341 ($7 orders).
- **Operational Thresholds:** Prep time mean = $27.37 mins; Delivery time mean = $24.16 mins.
- **Cuisine Leader:** American (Top cuisine on weekends).

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APPENDIX

I have used the **full code version** and used the goggle colab to run the codes and generate the answers and graphs.

The codes are listed below;

```
import pandas as pd
import numpy as np I
import matplotlib.pyplot as plt
import seaborn as sns
 from google.colab import files
# --- STEP 0: UPLOAD THE FILE
--- print("Please select your 'foodhub_order' CSV file:")
uploaded = files.upload()
filename = list(uploaded.keys())[0]
 df = pd.read_csv(filename)
print(f"Successfully loaded {filename}!")
```

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QUESTION 1 & 2:

- df.info()
- print(f"Rows: {df.shape[0]},
- Columns: {df.shape[1]}") print(df.info())
- QUESTION 3:
- # Check for missing values: df.isnull().sum()
- QUESTION 4:
- Get statistical summary df['food_preparation_time'].describe()
- Question 5:
- # Count 'Not given' entries df[df['rating'] == 'Not given'].shape[0]
- Question 6:
- # Cost of Order Distribution sns.histplot(data=df, x='cost_of_the_order', kde=True) plt.show() # Cuisine Type Distribution plt.figure(figsize=(15,5)) sns.countplot(data=df, x='cuisine_type') plt.xticks(rotation=45) plt.show()

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- Question 7:
- # Top 5 restaurants
- df['restaurant_name'].value_counts().head(5)
- Question 8:
- # Filter by weekend and find the most frequent cuisine
- df[df['day_of_the_week'] == 'Weekend']['cuisine_type'].value_counts().idxmax()
- Question 9:
- # Calculate percentage
- (df[df['cost_of_the_order'] > 20].shape[0] / df.shape[0]) * 100
- Question 10:
- # Average delivery time: df['delivery_time'].mean()
- Question 11:
- # Top 5 customer IDs: df['customer_id'].value_counts().head(5)

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- Question 12:
- # Correlation Heatmap:
- sns.heatmap:(df.corr(numeric_only=True), annot=True, cmap='coolwarm') plt.show()
- # Cuisine vs Cost:
- plt.figure(figsize=(15,7)) sns.boxplot(data=df, x='cuisine_type', y='cost_of_the_order') plt.xticks(rotation=45) plt.show()
- Question 13:
- # Filter and group:
- df_rated = df[df['rating'] != 'Not given'].copy()
- df_rated['rating'] = df_rated['rating'].astype(int)
- stats = df_rated.groupby('restaurant_name')['rating'].agg(['count', 'mean'])
- stats[(stats['count'] > 50) & (stats['mean'] > 4)]

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- Question 14:
- Calculate revenue: 25% if > $20, 15% if > $5
- def get_rev(x): if x > 20: return x * 0.25 elif x > 5:
- return x * 0.15 return 0 df['revenue'] = df['cost_of_the_order'].apply(get_rev) df['revenue'].sum()
- Question 15:
- # Total time = prep time + delivery time
- df['total_time'] = df['food_preparation_time'] + df['delivery_time']
- (df[df['total_time'] > 60].shape[0] / df.shape[0]) * 100
- Question 16:
- # Mean delivery time by day type:
- df.groupby('day_of_the_week')['delivery_time'].mean()

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